

TANNAS, Sweden

WMO: 023080

Lat: 62.449N

Lon: 12.668E

Elev: 2375

StdP: 13.48

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.3	(c) -1.3	(d) -11.2	(e) 3.3	(f) -5.6	(g) -6.0	(h) 4.4	(i) -0.4	(j) 13.5	(k) 30.9	(l) 11.8	(m) 29.7	(n) 0.7	(o) 120	(p) 0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.1	(c) 73.3	(d) 58.6	(e) 69.5	(f) 56.7	(g) 66.0	(h) 55.1	(i) 60.7	(j) 69.2	(k) 58.7	(l) 65.8	(m) 56.8	(n) 63.5	(o) 3.5	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.5	77.1	62.9	55.6	71.8	61.2	53.8	67.2	59.5	27.9	69.2	26.5	66.1	25.3	63.6	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.7	(b) 9.1	(c) 7.6	(e) -10.6	(f) 77.9	(g) 5.0	(h) 3.5	(i) -14.2	(j) 80.4	(k) -17.1	(l) 82.5	(m) -19.9	(n) 84.4	(o) -23.5	(p) 87.0		
(4) 10.7	(5) 9.1	(6) 7.6	(7) -11.2	(8) 63.7	(9) 4.8	(10) 2.2	(11) -14.7	(12) 65.3	(13) -17.5	(14) 66.6	(15) -20.3	(16) 67.8	(17) -23.8	(18) 69.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.3	18.9	20.1	24.4	33.1	42.1	49.3	54.9	52.5	45.3	35.0	25.9	20.7
	DBStd	14.81	9.23	9.56	8.16	6.48	7.55	6.63	5.96	5.68	5.79	6.48	8.04	9.69
	HDD50	5782	964	837	792	508	269	93	20	39	162	466	724	908
	HDD65	10859	1429	1257	1257	958	710	473	319	389	590	930	1174	1373
	CDD50	404	0	0	0	0	24	71	171	115	22	1	0	0
	CDD65	5	0	0	0	0	0	0	4	1	0	0	0	0
	CDH74	77	0	0	0	0	5	13	52	7	0	0	0	0
	CDH80	4	0	0	0	0	0	0	4	0	0	0	0	0
Wind	WSAvg	3.1	3.2	3.3	3.7	3.4	3.5	3.1	2.6	2.5	2.9	3.2	3.0	3.0
Precipitation	PrecAvg	24.7	1.7	1.2	1.2	1.3	2.1	2.9	3.5	3.5	2.0	2.0	1.9	1.4
	PrecMax	30.3	3.5	2.1	1.9	3.3	4.2	5.3	6.6	5.4	3.5	5.0	4.5	2.2
	PrecMin	19.3	0.4	0.2	0.4	0.2	0.6	0.7	1.1	1.6	0.6	0.6	0.8	0.4
	PrecStd	2.9	0.8	0.4	0.4	0.7	1.0	1.2	1.3	1.2	0.9	1.2	0.8	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.0	39.6	44.7	55.2	73.5	75.9	79.7	74.7	66.8	54.9	45.8	39.8
		MCWB	34.5	35.0	37.2	44.0	55.4	56.9	62.7	59.8	54.3	48.0	43.4	37.5
	2%	DB	35.7	36.8	41.0	50.3	66.6	70.4	74.4	70.3	61.6	49.8	41.6	37.5
		MCWB	33.4	33.5	35.9	40.9	51.8	54.8	60.3	57.7	53.0	45.5	39.5	35.4
	5%	DB	33.5	34.7	37.8	46.3	60.9	65.8	70.3	66.4	57.3	46.9	38.0	35.2
		MCWB	31.7	32.0	33.6	38.9	49.0	52.8	58.4	56.6	51.0	43.9	36.5	33.5
	10%	DB	31.1	32.3	35.4	42.7	55.1	61.5	66.4	62.5	54.1	44.2	35.6	33.0
		MCWB	29.6	30.2	32.2	36.9	45.6	51.4	56.7	55.2	49.6	41.7	34.2	31.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.2	36.3	38.6	46.0	56.4	59.7	64.3	62.1	56.6	50.0	43.4	38.2
		MCDB	37.1	38.8	42.9	53.0	68.8	68.5	75.0	70.5	62.7	52.5	45.3	39.4
	2%	WB	33.6	34.1	36.2	41.9	53.0	56.5	61.9	59.7	54.2	46.5	39.7	35.6
		MCDB	35.2	36.1	40.1	48.5	64.1	66.8	71.4	66.3	59.3	48.9	41.4	37.0
	5%	WB	31.8	32.2	34.2	39.5	49.8	54.5	59.7	57.9	52.1	44.1	36.7	33.7
		MCDB	33.3	34.5	37.5	45.0	58.7	63.6	67.9	64.1	56.4	46.3	37.8	35.1
	10%	WB	29.6	30.1	32.2	37.6	46.8	52.4	57.8	56.0	50.1	42.0	34.3	31.5
		MCDB	31.1	32.4	35.1	42.1	54.3	60.1	64.7	61.6	53.4	44.1	35.3	32.9
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.5	13.0	13.1	16.5	16.9	17.1	16.2	13.7	10.0	7.8	8.8
		MCDBR	9.8	10.2	13.5	17.5	25.7	24.3	24.4	22.4	19.3	14.6	9.5	9.0
		MCWBR	8.8	8.4	9.5	10.3	13.2	12.0	12.2	11.8	11.3	10.2	8.5	8.5
	5% WB	MCDBR	9.3	9.6	12.5	15.8	23.4	20.7	21.3	19.0	17.2	12.5	9.2	9.0
		MCWBR	8.6	8.3	9.2	9.7	12.9	11.1	11.4	11.0	10.8	9.4	8.5	8.8
Clear-Sky Solar Irradiance	taub	0.194	0.250	0.279	0.311	0.323	0.319	0.332	0.322	0.294	0.277	0.207	0.170	
	taud	2.065	2.249	2.260	2.314	2.466	2.516	2.487	2.504	2.567	2.479	2.134	1.971	
	Ebn at Noon	162	231	265	279	285	288	280	274	264	223	153	103	
	Edh at Noon	10	19	27	32	30	30	30	27	21	15	9	6	
All-Sky Solar Radiation	RadAvg	73	267	738	1322	1510	1610	1474	1140	727	316	97	38	
	RadStd	7	43	62	111	111	139	177	104	81	47	17	3	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page